Design, Fabrication and Experimental Analysis of Unbalanced Silicon Photonic Mach–Zehnder Interferometers

Arefur Rahman

I. Introduction

Mach-Zehnder interferometers (MZIs) are fundamental building blocks in integrated photonics, serving critical roles in optical filtering, modulation, switching, and sensing. Their utility stems from their acute sensitivity to optical phase differences between two parallel waveguide arms, rendering them highly effective for wavelength-selective routing and on-chip signal processing within silicon photonic platforms. While the underlying theory governing MZI operation is well established, optimizing their physical implementation requires a rigorous understanding of how specific component architectures and inherent fabrication tolerances collectively influence practical optical performance.

This work presents a comprehensive study encompassing the design, fabrication, and experimental characterization of unbalanced silicon photonic MZIs. Three distinct interferometer topologies are systematically investigated: (i) standard configurations employing Y-branches for both beam splitting and recombination, (ii) hybrid structures utilizing a Y-branch splitter paired with a broadband directional coupler, and (iii) advanced architectures incorporating a ring resonator within one interferometer arm. By varying the differential path length (ΔL) across these designs, the resulting effects on the transmission spectrum and free spectral range (FSR) are examined for both transverse-electric (TE) and transverse-magnetic (TM) polarization modes.

Furthermore, to reconcile discrepancies between idealized device simulations and physical fabrication outcomes, a corner analysis methodology is employed to quantify the impact of lithographic dimensional variations on the waveguide group indices. The designed photonic circuits were fabricated using a 100 keV electron-beam lithography (EBL) process. Experimental characterization of the fabricated devices demonstrates excellent quantitative agreement between measured results and the corresponding theoretical and simulation-based predictions.

II. Theory

A. Transfer Function of a Mach–Zehnder Interferometer

The Mach–Zehnder interferometer (MZI) consists of an input optical signal that is split into two branches and subsequently recombined. The splitting and combining operations can be implemented using various optical components, including Y-branches and directional couplers. In this design proposal, we consider two interferometer implementations: (i) an MZI formed using two Y-branches and (ii) an MZI formed using a Y-branch as the splitter and a broadband directional coupler as the combiner.

Let the input optical intensity be I_i, corresponding to an input electric field E_i. At the output of the splitter, the field is equally divided between the two arms such that

$$E_1 = \frac{E_i}{\sqrt{2}}, \qquad E_2 = \frac{E_i}{\sqrt{2}}. \tag{1}$$

The propagation of light in the two waveguides is described by the propagation constants

$$\beta_1 = \frac{2\pi n_1}{\lambda}, \qquad \beta_2 = \frac{2\pi n_2}{\lambda}, \tag{2}$$

where n_1 and n_2 are the effective refractive indices of the two waveguides, and λ is the vacuum wavelength. The waveguide lengths are L_1 and $L_2 = L_1 + \Delta L$, with corresponding power attenuation coefficients α_1 and α_2.

At the inputs of the combiner, the optical fields are

$$E_{o1} = \frac{E_i}{\sqrt{2}} e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1}, \tag{3}$$

$$E_{o2} = \frac{E_i}{\sqrt{2}} e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}. \tag{4}$$

Two Y-branch MZI: When the combiner is implemented using a second Y-branch, the output field is given by

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + E_{o2}) = \frac{E_i}{2}\left(e^{-i\phi_1} + e^{-i\phi_2}\right), \tag{5}$$

where $\phi_k = \beta_k L_k$. The corresponding output intensity is

$$I_o = \frac{I_i}{4}\left|e^{-i\phi_1} + e^{-i\phi_2}\right|^2. \tag{6}$$

Neglecting propagation loss ($\alpha_1 = \alpha_2 = 0$), this simplifies to the well-known MZI transfer function

$$I_o = \frac{I_i}{2}\left[1 + \cos(\phi_1 - \phi_2)\right]. \tag{7}$$

Y-branch and directional-coupler MZI: If the output combiner is instead implemented using a broadband directional coupler, the recombination introduces an additional $\pi/2$ relative phase shift between the two inputs. For an ideal $50/50$ directional coupler, the output field can be written as

Email: rahma221@purdue.edu

$$E_o = \frac{1}{\sqrt{2}}(E_{o1} + iE_{o2}) = \frac{E_i}{2}\left(e^{-i\phi_1} + ie^{-i\phi_2}\right). \quad (8)$$

The resulting output intensity is

$$I_o = \frac{I_i}{4}\left|e^{-i\phi_1} + ie^{-i\phi_2}\right|^2, \quad (9)$$

which, in the absence of loss, simplifies to

$$I_o = \frac{I_i}{2}\left[1 + \sin(\phi_1 - \phi_2)\right]. \quad (10)$$

Thus, replacing the output Y-branch with a directional coupler introduces a constant $\pi/2$ phase offset in the interferometer response. Both implementations are equivalent up to this fixed phase bias and exhibit the same free spectral range, which is determined solely by the optical path-length difference ΔL.

B. Free Spectral Range of an Imbalanced Interferometer

Given the transfer function of an imbalanced Mach–Zehnder interferometer in 10, we seek to determine the spacing between adjacent transmission maxima, referred to as the free spectral range (FSR),

$$\mathrm{FSR} = \Delta\lambda = \lambda_{m+1} - \lambda_m. \quad (11)$$

The phase term in the interferometer transfer function is defined as

$$\delta = \beta\Delta L, \quad (12)$$

where the propagation constant is

$$\beta = \frac{2\pi n}{\lambda}, \quad (13)$$

with n denoting the effective refractive index and λ the vacuum wavelength. Adjacent transmission peaks correspond to a phase difference of 2π, such that

$$\Delta\delta = \delta_m - \delta_{m+1} = 2\pi. \quad (14)$$

In typical interferometers, many fringes are observed across the spectrum, implying operation at high interference order ($\delta \gg 2\pi$). Writing the phase difference explicitly in terms of the propagation constants at adjacent peaks yields

$$2\pi = \delta_m - \delta_{m+1} = \beta_m\Delta L - \beta_{m+1}\Delta L. \quad (15)$$

This gives the corresponding change in propagation constant,

$$\Delta\beta = \beta_m - \beta_{m+1} = \frac{2\pi}{\Delta L}. \quad (16)$$

Assuming that β varies approximately linearly with wavelength over the interval of interest, we write

$$\Delta\beta \approx -\frac{d\beta}{d\lambda}\Delta\lambda. \quad (17)$$

Combining the above expressions leads to

$$\Delta\lambda \approx -\frac{2\pi}{\Delta L}\left(\frac{d\beta}{d\lambda}\right)^{-1}. \quad (18)$$

To evaluate $d\beta/d\lambda$, we differentiate $\beta = 2\pi n/\lambda$ with respect to λ,

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda}\frac{dn}{d\lambda} - \frac{2\pi n}{\lambda^2} = -\frac{2\pi}{\lambda^2}\left(n - \lambda\frac{dn}{d\lambda}\right). \quad (19)$$

Substituting this result into the expression for $\Delta\lambda$ yields the free spectral range,

$$\mathrm{FSR} = \Delta\lambda = \frac{\lambda^2}{\Delta L\left(n - \lambda\frac{dn}{d\lambda}\right)}. \quad (20)$$

Defining the group index n_g as

$$n_g = n - \lambda\frac{dn}{d\lambda}, \quad (21)$$

the free spectral range can be written in its commonly used form,

$$\mathrm{FSR} = \frac{\lambda^2}{\Delta L\, n_g}. \quad (22)$$

III. Modelling and Simulation

A. Waveguide

The fundamental optical properties of the waveguide were characterized using the eigenmode solver in Lumerical MODE. The simulated structure consists of a standard silicon strip waveguide with a 500 nm core width and 220 nm thickness, fully clad in silicon dioxide. Eigenmode simulations were performed across the 1500–1600 nm wavelength range for both the fundamental TE and TM modes to verify single-mode operation and assess dispersive behavior.

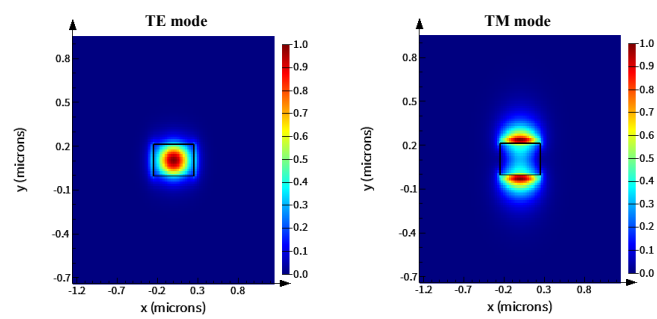

Fig. 1. **Simulated mode profiles**.

From these simulations, the wavelength-dependent effective index (n_{eff}) and group index (n_g) were extracted. For efficient circuit-level modeling, the effective index was approximated by a second-order Taylor expansion about a reference wavelength λ_0. The fitted coefficients n_1, n_2, and n_3 are summarized in Table I. This compact analytical model reproduces the numerical eigenmode results with high fidelity, enabling accurate simulation of the interferometric circuits.

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (23)$$

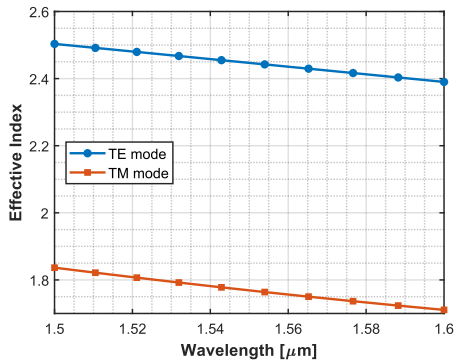

Fig. 2. **Simulated fundamental modes effective index**.

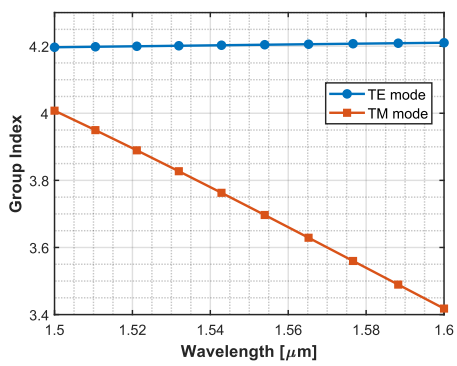

Fig. 3. **Simulated fundamental modes group index**.

TABLE I
WAVEGUIDE COMPACT MODEL COEFFICIENTS

Coefficient	TE Value	TM Value
n_1	2.446	1.769
n_2	-1.133	-1.259
n_3	-0.044	1.913

B. MZI Circuit

Following the component-level analysis, system-level simulations of nine unbalanced Mach-Zehnder Interferometers (MZI1–MZI9, Fig. 11) were conducted in Lumerical INTERCONNECT. Circuit topologies were constructed using EBeam compact models from the SiEPIC-EBeam-PDK library, with grating couplers serving as the optical input and output ports. Interconnecting waveguides were parameterized using data exported from the preceding modal analysis, while all remaining passive components—including Y-branch splitters, broadband directional couplers, and ring resonators—were characterized via S-parameters sourced directly from the PDK library.

Three distinct architectural configurations were evaluated:

- Dual Y-branch designs (MZI1–MZI4), in which both the splitter and combiner are implemented as Y-branch junctions.
- **Hybrid designs** employing a Y-branch splitter paired with a broadband directional coupler as the output combiner (MZI5–MZI7).
- Ring-integrated designs in which a ring resonator is embedded within one arm of the interferometer (MZI8–MZI9).

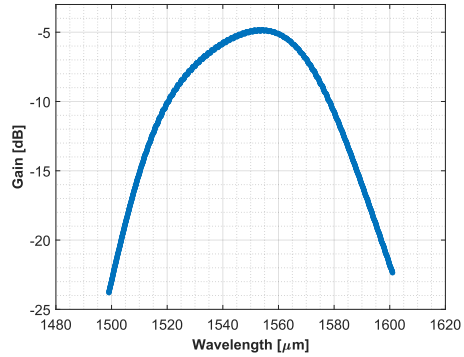

Fig. 4. Insertion loss reference spectrum of grating coupler.

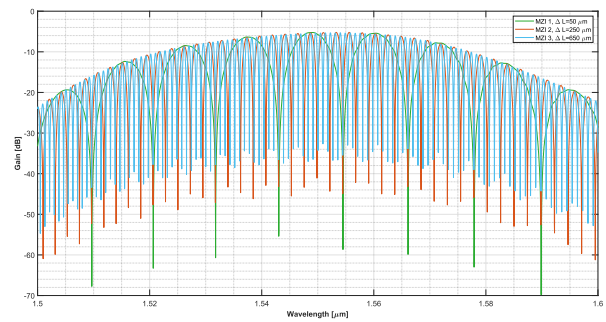

Fig. 5. Simulated optical gain as a function of wavelength for three dual Y-branch unbalanced Mach-Zehnder Interferometers (MZI1, MZI2, and MZI3). The devices are configured with path-length differences (ΔL) of 50 µm, 250 µm, and 650 µm, respectively.

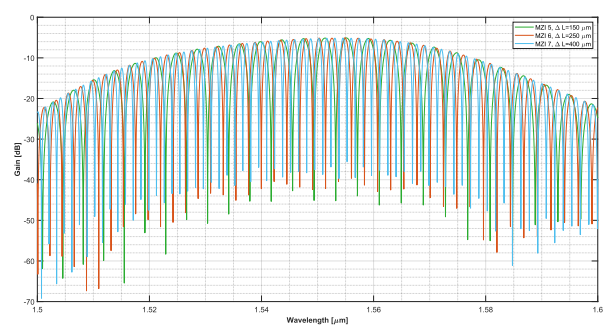

Fig. 6. Simulated optical gain as a function of wavelength for three unbalanced Mach-Zehnder Interferometers (MZI5, MZI6, and MZI7). Each device features a Y-branch splitter and a broadband directional coupler. The path-length differences (ΔL) are 150 µm, 250 µm, and 400 µm, respectively.

Device performance was assessed through spectral sweeps over the $1500 - -1600$ nm wavelength range. Grating coupler insertion-loss contributions are presented in Fig. 4. The resulting simulated transfer functions corroborated several foundational theoretical predictions:

- *Path-Length Dependence:* Increasing the physical arm-length mismatch (ΔL) proportionally increases the frequency of spectral oscillations, thereby reducing the Free Spectral Range (FSR) , shown in Fig. 5 and Fig 6,.

- *Polarization Dependence:* For a fixed ΔL, the TE and TM modes exhibit distinct FSRs owing to their respective polarization-dependent group indices, as illustrated in Fig. 7 between MZI2 (TE) and MZI4 (TM).
- *Combiner Phase Offset:* Substituting the output Y-branch with a broadband directional coupler (as observed in comparing MZI2 with MZI6 in Fig. 7) preserves the FSR while introducing a relative phase offset, effectively translating the interference fringes along the wavelength axis.
- *Ring Resonator Integration:* Incorporating a ring resonator fundamentally modifies the device's transmission profile (Fig. 8) [1]. When the optical signal is routed through the ring's *through-port* (MZI8), sharp phase discontinuities and amplitude distortions are introduced at resonant wavelengths, while conventional two-beam interference is maintained off-resonance. Conversely, routing the signal via the drop port (MZI9) imposes a narrow-bandpass response that suppresses broadband interference, confining optical transmission to the ring's resonant wavelengths.

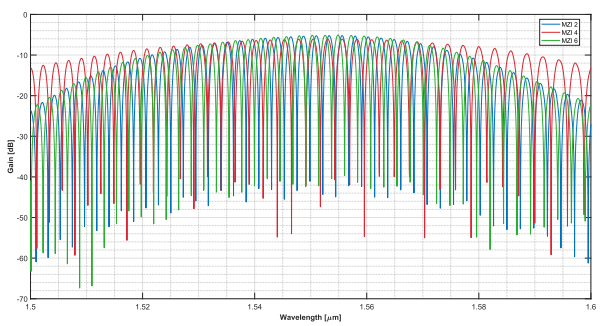

Fig. 7. Simulated optical gain as a function of wavelength for three unbalanced Mach-Zehnder Interferometers (MZI2, MZI4, and MZI6), each with a fixed path-length difference of 250 µm. MZI2 (TE mode) and MZI4 (TM mode) employ a dual Y-branch architecture, whereas MZI6 (TE mode) features a Y-branch splitter and a directional coupler.

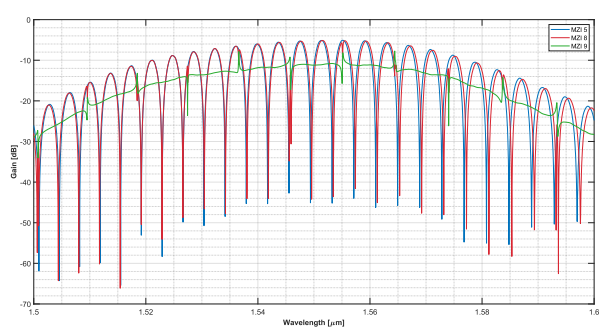

Fig. 8. Simulated optical gain as a function of wavelength for three unbalanced Mach-Zehnder Interferometers (MZI5, MZI8, and MZI9), each with a fixed path-length difference of 150 µm and an architecture comprising a Y-branch splitter and a directional coupler. MZI5 represents the baseline configuration without a ring resonator. MZI8 and MZI9 integrate a ring resonator into one of the interferometer arms, with the optical signal routed via the ring's through port and drop port, respectively.

IV. Fabrication

A. Fabrication Process

The photonic devices were fabricated using the NanoSOI multi-project wafer (MPW) process at Applied Nanotools Inc. (Edmonton, Canada) [8]. This process utilizes direct-write 100 keV electron beam lithography (EBL). Silicon-on-insulator (SOI) wafers with a 200 mm diameter, a 220 nm device layer thickness, and a 2 μm buffer oxide thickness served as the base material. The fabrication proceeded according to the following steps:

1) **Substrate Preparation:** The SOI wafer was pre-diced into 25×25 mm square substrates. Lines were scribed into the backsides of the substrates to facilitate easy separation into individual chips upon completion.
2) **Cleaning and Resist Coating:** The substrates underwent an initial 15-minute clean in piranha solution (3:1 H_2SO_4 : H_2O_2), followed by a water and isopropyl alcohol (IPA) rinse. A hydrogen silsesquioxane (HSQ) resist was then spin-coated onto the substrates and baked to evaporate the solvent.
3) **Lithographic Patterning:** Device layouts were patterned using a JEOL JBX-8100FS electron beam instrument at the University of British Columbia. The exposure dosage was rigorously corrected for proximity effects caused by electron backscatter, and the shape writing order was optimized for efficient patterning and minimized beam drift.
4) **Development:** Following EBL exposure, the resist was developed in a tetramethylammonium hydroxide (TMAH) solution. The devices were subsequently inspected optically to identify any residues or defects.
5) **Etching:** The chips were mounted onto a 4-inch handle wafer and subjected to an anisotropic inductively coupled plasma reactive ion etching (ICP-RIE) process using chlorine, strictly performed after qualifying the etch rate.
6) **Resist Stripping and SEM Inspection:** The remaining HSQ resist was removed using a 10:1 buffered oxide wet etch. The structures were then evaluated using a scanning electron microscope (SEM) to verify overall patterning fidelity and etch quality.
7) **Cladding Deposition:** A 2.2 μm upper oxide cladding was deposited at 300°C via a plasma-enhanced chemical vapor deposition (PECVD) process utilizing tetraethyl orthosilicate (TEOS).
8) **Metrology:** Final reflectrometry measurements were conducted to confirm the exact thicknesses of the device layer, buffer oxide, and upper cladding prior to chip delivery.

B. Manufacturing Variability

Fabrication processes inherently introduce dimensional variations that can significantly impact device performance. Robust photonic design must therefore account for such structural deviations, which arise from sources including lithographic smoothing, fluctuations in resist properties (e.g., thickness,

sensitivity, and aging), and variability in exposure, development, and etching conditions.

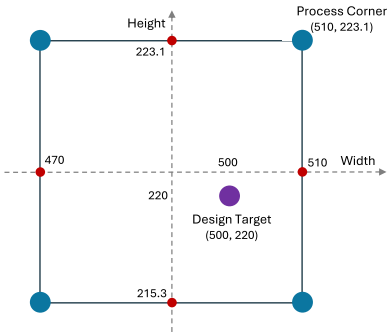

Fig. 9. Waveguide corner profiles resulting from cross-sectional variation introduced during fabrication.

To illustrate this, fabrication data from the Washington University Nanofabrication Facility were examined for silicon waveguides with nominal dimensions of 500 nm (width) $\times$220 nm (thickness). The measured widths ranged from 470 nm to 510 nm, and the thicknesses from 215.3 nm to 223.1 nm. Based on these empirical tolerances, a corner analysis was performed to characterize the four extreme cross-sectional configurations, as illustrated in Fig 9.

TABLE II
GROUP INDEX VS. WAVEGUIDE CROSS-SECTION AT 1550 NM (CORNER ANALYSIS).

Cross-Section (nm^2)	**Group Index** (n_g)	
	TE Mode	**TM Mode**
500×220	4.20	3.72
470×223.1	4.26	3.74
510×223.1	4.19	3.84
510×215.3	4.18	3.59
470×215.3	4.25	3.48

The results of this corner analysis, summarized in Table II, reveal the substantial influence of fabrication-induced structural variations on the waveguide's optical properties at a wavelength of 1550 nm. For the nominal 500×220 nm^2 cross-section, the group index of the TE mode spans from 4.18 to 4.26, while that of the fundamental quasi-TM mode ranges from 3.48 to 3.84.

V. EXPERIMENT

A. Measurement Description

To characterize the devices, a custom-built automated test setup ([3], [7]) with automated control software written in Python was used [4]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [5]. A 90° rotation was used to inject light into the TM grating couplers [5]. A polarization maintaining fibre array was used to couple light in/out of the chip [6].

B. Experimental Results

To evaluate the performance of the fabricated devices, key waveguide parameters were extracted directly from the experimental transmission spectra. The raw data were first truncated to a target wavelength range (1526–1574 nm) and subjected to baseline correction via a polynomial fit, which was used to approximate and subtract the transfer function envelope of the grating couplers. This normalization approach, shown in Fig. 10, proved sufficiently accurate that data from dedicated de-embedding test structures (DE1, Fig 11) were not required for further calibration.

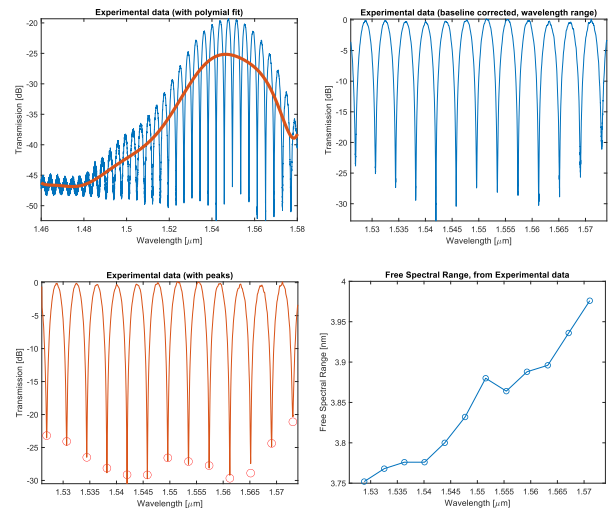

Fig. 10. Overview of the experimental data analysis pipeline.

Following baseline correction, the spectra were smoothed using a moving-average filter. The FSR of each MZI was subsequently determined by identifying the wavelength spacing between adjacent transmission minima via a peak-finding algorithm. The corresponding waveguide group indices were then derived analytically from the measured FSR values.

The experimentally extracted group indices fall within the predicted bounds established by the preceding corner analysis. Table III presents a systematic comparison of simulated and experimental results over various device structures, path-length differences, and polarization modes, all evaluated near a central wavelength of 1.55μm. The data demonstrate strong quantitative consistency between the theoretical estimates and the measured device characteristics, supporting the accuracy of the simulation framework.

VI. ACKNOWLEDGEMENT

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TABLE III
COMPARISON OF SIMULATION AND EXPERIMENTAL RESULTS

Circuit Name	Mode	ΔL (μm)	Simulation		Experiment			
					Scylla Dataset		Iris Dataset	
			Group Index	FSR (nm)	Group Index	FSR (nm)	Group Index	FSR (nm)
MZI1	TE	50	4.20	11.44	4.19	11.47	4.21	11.45
MZI2	TE	250	4.20	2.29	4.19	2.29	4.22	2.30
MZI3	TE	650	4.20	0.88	4.19	0.89	4.19	0.89
MZI4	TM	250	3.72	2.60	3.89	2.57	-	-
MZI5	TE	150	4.20	3.81	4.16	3.86	4.16	3.86
MZI6	TE	250	4.20	2.29	4.18	2.30	4.20	2.31
MZI7	TE	400	4.20	1.43	4.20	1.43	4.18	1.44
MZI8	TE	150	4.20	3.81	4.29	3.86	4.28	3.87

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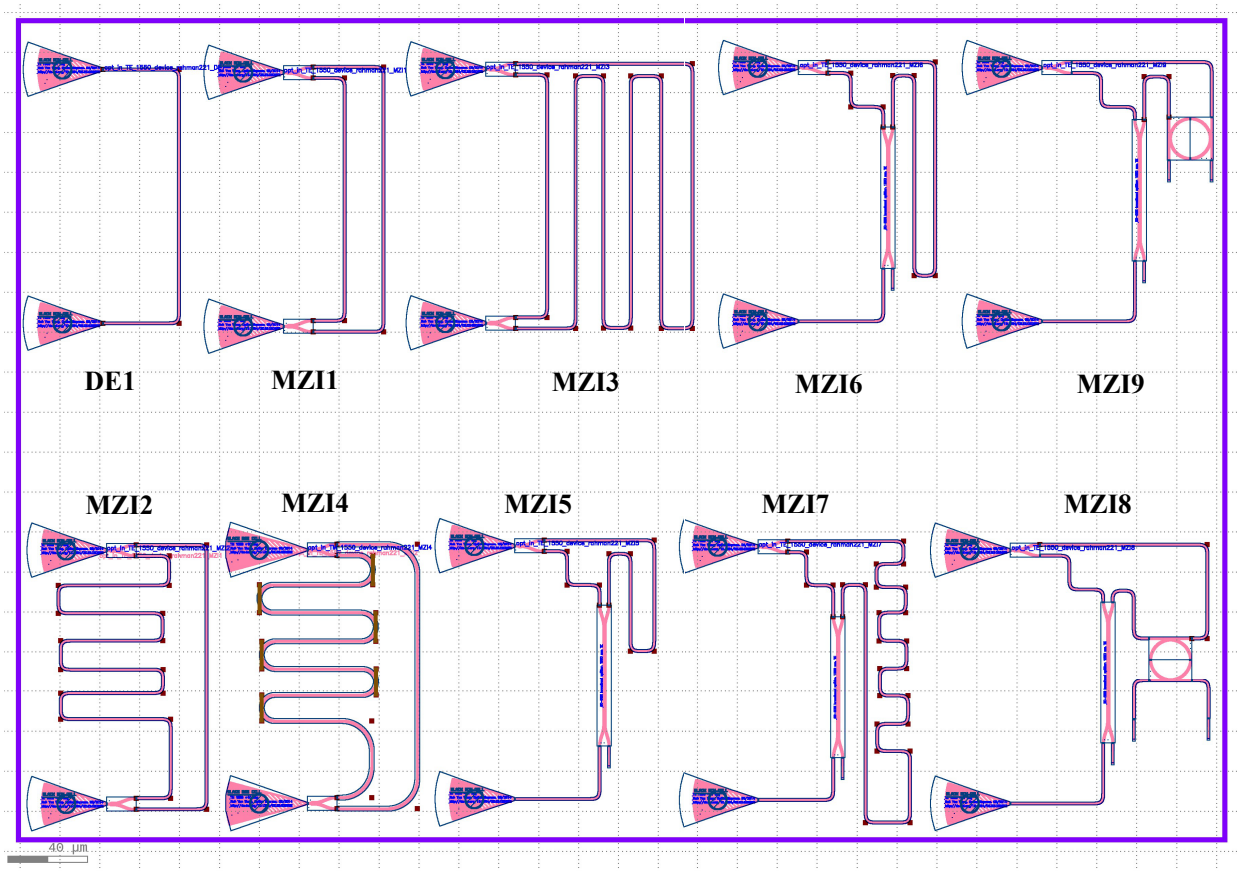

Fig. 11. GDS layout of the designed devices.